User Manual

for the EEVblog 121GW multimeter with the open firmware F-1.24



This manual shows you how to use your 121GW with the open firmware: every key, every measurement, every setting and every new feature, step by step. The pictures of the display were made by running this firmware in the 121GW emulator, so they show the same texts as your meter.

The open firmware is not made or supported by EEVblog. You install it at your own risk (see the licence).

Contents

What's new in this firmware 3

Safety first 4

Your meter 4
- The dial 4
- The keys 5
- The display 5

Quick start 6
- Batteries 6
- Switch on 6
- Your first measurement 6

Measuring 7
- Voltage 7
- Temperature 7
- Frequency, pulse, duty cycle 7
- Resistance, continuity, diode, capacitance 7
- Current 8
- Power (VA) 8
- Ranges and OFL 8
- Two keys together 8

Everyday functions 9
- HOLD and auto-hold 9
- REL 9
- MIN / MAX 9
- PEAK and noise filter 9
- Backlight 9
- Zero offset for currents 10

The new features 10
- Pass / fail testing (limit test) 10
- REL in percent 10
- Statistics in MIN / MAX 11
- dBm with your reference 11
- Six thermocouple types 12
- PT100 / PT1000 sensors 12
- 4–20 mA loop 12
- Better continuity 13
- Faster digits (display level) 13
- Steady capacitance range (anti-hunting) 13
- Diode contact chirp 14
- Backlight timer 14
- Faster Bluetooth 14
- 12-hour clock 14
- Smart logging (trigger) 15
- Battery check and trim 15
- Quiet mode that is really quiet 15
- Faster measuring 15
- Big SD cards and formatting 16
- Bluetooth file transfer 16
- Extended link diagnosis 16
- Protected calibration backup 17

The settings menu (SETUP) 17
- Changing a setting 17
- All pages, in order 18

Logging and the SD card 19
- Record readings 19
- The files 20
- Look at the last readings 20
- Cards 20

Bluetooth and the Companion app 20

Backup of your calibration 22
- Make a backup 22
- Restore it 22
- Serial number 22
- Keys while switching on 22

Updating the firmware 23
- The easy way: Companion app 23
- With an SD card 23
- The two-part bootloader 24

Messages 24

Troubleshooting 25

Measuring ranges 25

What's new in this firmware

The open firmware keeps everything you know from the original 121GW and adds a lot. Here is what you get. Click a page number to jump there.

Feature	What it does for you	
Pass / fail testing	Sort parts as PASS, too high or too low, with a beep	page 10
REL in percent	See how far a part is from a reference, in %	page 10
Statistics	Count, standard deviation, time of the max and min	page 11
dBm with your reference	Audio levels for 600 Ω, 50 Ω, 75 Ω or dBV	page 11
Six thermocouple types	K, J, T, N, R and S	page 12
PT100 / PT1000	Read a platinum sensor directly as a temperature	page 12
4–20 mA loop	See a loop current as 0–100 %	page 12
Better continuity	Beep below or above 1–300 Ω, adjustable beep hold	page 13
Faster digits	Choose calm or quick-reacting digits	page 13
Steady capacitance range	Stops the range hunting with some 100 nF capacitors	page 13
Backlight timer	The light switches off by itself to save the batteries	page 14
Faster Bluetooth	Up to 20 readings per second to your phone or computer	page 14
12-hour clock	Time shown as 1–12	page 14
Smart logging	Log only on a change or a pass/fail change	page 15
Battery check	Correct battery voltage reading, adjustable	page 15
Quiet mode that is really quiet	Buzzer off silences key and info beeps	page 15
Faster measuring	Quicker display updates	page 15
Big SD cards	exFAT cards, long file names, format on the meter	page 16
Bluetooth file transfer	Firmware, log files and settings over Bluetooth	page 16
Protected calibration backup	A backup can only go back into its own meter	page 17

Feature	What it does for you	
Link diagnosis	Error counts of the Bluetooth link, for problem reports	page 16
Companion app	Live view, scope, recording, updates, card files, settings	page 20

> **NOTE** Your calibration, serial number and settings stay in the meter when you install this firmware. If you ever go back to the original firmware, they are still there.

Safety first

> **WARNING** The meter is rated **600 V CAT III**. Never measure above that. Use only test leads and accessories rated at least as high as the meter.

- Take the leads off the circuit **before you turn the dial**.
- Check that the leads are in the **right sockets** before you measure.
- **Check your leads** before use. Replace damaged leads.
- Use the meter **indoors and dry** only.
- The **A** socket is for up to 10 A, the **mA/µA** socket for up to 400 mA. Both have a fuse. If a lead is in a current socket while the dial is on something else, the meter shows `LEAd` and beeps. Move the lead first.
- From about 30 V the **lightning symbol** lights up as a reminder.
- **Discharge capacitors** before you measure them.
- Before you **open the meter** (batteries, fuses), take the leads off and switch it off. Replace fuses only with the same type: 400 mA / 600 V (6×32 mm) for mA/µA, 11 A / 1000 V (10×38 mm) for A.
- **Fuse test:** dial on a voltage function, connect a lead from V Ω to the A (or mA) socket. A beep = fuse OK.
- **AC with a large DC part:** auto-range can choose the wrong range. Use manual range (RANGE key).

Your meter

The dial

The dial switches the meter on and chooses what you measure. Both ends are **OFF**. Press **MODE** to switch between the functions of a position. Each position remembers your last choice, also after switching off.

Position	Functions (MODE)
Low Z	Voltage with low input resistance
V	DC → AC → AC+DC volts
mV / Temp	DC mV → AC mV → temperature
Hz ms %	frequency → pulse width → duty cycle
Ω	resistance → continuity → diode → capacitance
mVA / VA	power (volts × amps)
µA / µVA	small currents (µA) and small power (µVA)
A / mA	AC ↔ DC current (the socket you use picks mA or A)

The keys

Each key does one thing when you **press** it and often another when you **hold** it for half a second. One beep = done. Two short beeps = not possible right now.

Key	Press	Hold
MODE	Next function of the dial position	Backlight on / off
RANGE	Fix the range; press again for the next range. Diode: 3 V / 15 V	Back to auto-range
HOLD	Freeze the reading → auto-hold → off	–
REL	Show the difference to the present reading (REL) on / off	AC: 1 kHz filter on / off. Current: store a zero offset
PEAK	AC: catch short peaks on / off	Bluetooth on / off
MIN/MAX	Start; then show max → min → average → all	Stop MIN/MAX
MEM	Show the last logged readings	Start / stop logging to the SD card
SETUP	Next page of the settings (small display)	Change the page shown; hold again to save

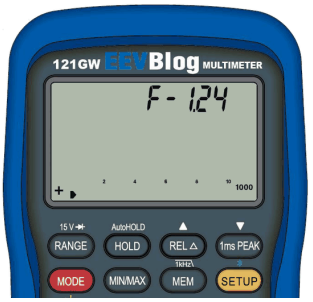
The display

The **big digits** show the reading. The **small display** above shows one page of the settings menu: normally the temperature inside the meter, or extra information for what you measure (for example the frequency of an AC voltage). The **bar graph** (the row of small bars) follows quick changes. **AUTO** means the meter picks the range (the size of the scale) by itself. Symbols show what is on: AUTO, HOLD, REL, PEAK, MIN MAX AVG, MEM, BT (Bluetooth), 1kHz, Apo (auto power-off), the battery and the lightning symbol for high voltage.

Quick start

Batteries

Four AA alkaline batteries (1.5 V). When the battery symbol shows, replace them. Lithium AA cells start with a higher voltage; EEVblog advises using them a little first. Take the batteries out if you store the meter for more than 50 days. A small CR1220 cell inside keeps the clock running.



The version at power-on.

Switch on

Turn the dial to a function. The small display shows the firmware version (`F-1.24`) for a moment, then the meter measures.

After 30 minutes without a key press the meter switches itself off (1, 15 or 60 minutes, or never: SETUP page `APO.30`). It beeps five times before; press any key to keep it on, or to switch it on again. It never switches off while logging or during MIN/MAX.



Measuring DC volts.

Your first measurement

1. Black lead into **COM**, red lead into **V Ω**.
2. Dial to **V**.
3. Touch the probes to the circuit. **AUTO** means the meter chooses the range for you.



The front of the meter.

Measuring

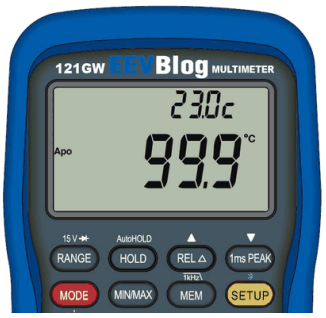
Always: dial first, then MODE for the function, then connect the leads.



AC volts, frequency on top



Low Z



Temperature



Frequency



Resistance



DC current

Voltage

V: up to 600 V, DC, AC or AC+DC. In AC the small display shows the frequency and, one page further, the level in dBm. **mV**: small voltages up to 500 mV. **Low Z**: makes "ghost" voltages from nearby wires disappear; it chooses AC or DC itself (dashes while it decides). Not for voltages below about 12 V.

Temperature

Dial to **mV / Temp**, press MODE twice, connect the thermocouple. K type is the default; other types page 12. °C or °F: first settings page.

Frequency, pulse, duty cycle

Dial to **Hz ms %**; MODE steps through Hz, ms and %.

Resistance, continuity, diode, capacitance

- **Resistance** up to 50 MΩ. Short the probes and press REL to subtract the lead resistance.
- **Continuity**: beeps below 30 Ω (you can change it page 13).

- **Diode**: shows the forward voltage. RANGE switches the test voltage between 3 V and 15 V. It goes back to 3 V after a mode change. Use 15 V only on parts that are off and discharged.
- **Capacitance** up to 10 000 µF. Discharge first; REL subtracts the leads. On the 100 nF range in auto range the meter waits up to **2.5 seconds** for a result before it goes on to the 1 µF range (the original firmware waits 2 seconds; 1.5 seconds on a meter with the board strap PF8 set), so a slow ceramic capacitor of about 100 nF can show its value with the full resolution. This is **experimental**: how long such a capacitor really takes is not measured yet. A capacitor that still keeps the range changing: **page 13**

Current

Put the red lead in **µA/mA** (small currents) or **A** (up to 10 A) first. On **A / mA** the meter sees which socket you use. MODE switches AC and DC.

Burden voltage (the voltage the meter itself drops): on the small display page `bd0FF` switch it to `bd0n` and connect a lead from the current socket you use to V Ω. In the current functions this choice is forgotten when you switch off.

Power (VA)

Measures voltage (V Ω socket) and current (mA or A socket) at the same time and shows the power, volts × amps. The small display shows the voltage or the current. Dashes until the first result. The burden page `bd` is here too; in the power functions its setting is **saved** and stays after switching off.

Ranges and OFL

RANGE fixes the range, each press goes one range up; hold it to go back to AUTO. `0FL` means the input is too big for the range.

Two keys together

Hold MEM and HOLD together: logging starts or stops, and HOLD, MIN/MAX and REL are switched off.

Everyday functions

HOLD and auto-hold



Auto-hold (A-HOLD).

Press **HOLD**: the reading freezes. Press again: **auto-hold** (A-HOLD). It waits until the reading is steady, keeps it with a beep and takes the next one when you measure again. In volts and amps it ignores readings near zero, so it waits while the probes are in the air. In resistance and diode it ignores open probes, and a short (0) counts. Press again: off.

In Hz, ms, %, capacitance and during MIN/MAX the second press switches HOLD off.

REL



REL shows the difference.

Press **REL**: the meter remembers the present reading and shows the difference to it. Press again: off. Example: measure a 100 Ω reference, press REL, then each other resistor shows how much it differs. Want it in percent? page 10

MIN / MAX



The maximum.

Press **MIN/MAX** to start. Each press shows: maximum, minimum, average, all three. Hold **MIN/MAX** to stop. While it runs you also get statistics page 11.

PEAK and noise filter

In AC volts and AC mV: **PEAK** catches short peaks (1 ms). Hold **REL** to switch the 1 kHz filter on or off; it lets through only frequencies below about 1 kHz. PEAK and the filter can't be on at the same time.

Backlight

Hold **MODE**. The meter remembers your choice. To let it switch off by itself page 14.

Zero offset for currents

In a current function on its lowest range, with nothing connected, hold REL. The small value shown is stored as the zero and subtracted from then on, also after switching off.

> **WARNING** Each time you do this it adds to the stored zero, and there is no key to clear it. Only use it when the meter really shows a current with nothing connected.

The new features

Each feature in a few steps. "Settings page" means the small display: press SETUP until the page shows, hold SETUP to change it (it blinks), REL / PEAK to choose, hold SETUP to save.

Pass / fail testing (limit test)

Where: any function · settings pages `Lo`, `HI`, `Lt`

Check many parts quickly: the meter says `PASS`, `HI` (too high) or `Lo` (too low).

1. Put a part with the **lowest allowed** value on the probes. Go to page `Lo`, hold SETUP, press REL (the value is taken), hold SETUP.
2. Same with the **highest allowed** value on page `HI`.
3. Page `Lt`: hold SETUP, REL for `LtOn`, hold SETUP.
4. Now test your parts. The small display shows the result. Two short beeps when a part leaves PASS, one beep when it is back in PASS.

Both limits must be taken in the same function and range (if Lo is above HI, the meter swaps them). While the test runs the range stays fixed. Turning the dial or changing the function ends it. The limits are forgotten when you switch off.



PASS



Too high

REL in percent

Where: any function · settings page `rEL`

Shows how far a part is from a reference, in percent. Great for checking tolerances.

1. Settings page `rEL-d`: change it to `rEL-P` and save.



10 % above the reference.

2. Measure the reference and press **REL**.
3. Measure the other parts. Example: reference 1.000 V, now 1.100 V → **10.000 %**.

Logging and Bluetooth still use the normal value.

Statistics in MIN / MAX

Where: while MIN/MAX runs · settings pages `n`, `Sd`, `tHI`, `tLo`

Start **MIN/MAX**, then press **SETUP** to the new pages:

- `n`: how many readings were taken.
- `Sd`: how much the reading wobbles around its average (the "standard deviation"), shown in the big digits in the unit of the reading. Small = steady.
- `tHI` / `tLo`: the **time** the maximum / minimum happened. Handy when you come back after an hour and want to know when the voltage dropped.

Log files also get the time of the max and min at the end.



Number of readings



Time of the maximum

dBm with your reference

Where: dial V, AC · the dBm page of the small display

In AC volts press **SETUP** until the level in dB shows. Hold **SETUP** to choose the reference: `r 600` (600 Ω), `r 50` (50 Ω), `r 75` (75 Ω) or `r dbV` (dB compared with 1 V). Hold SETUP to save.

Example: 1.000 V AC is 2.218 dBm at 600 Ω.

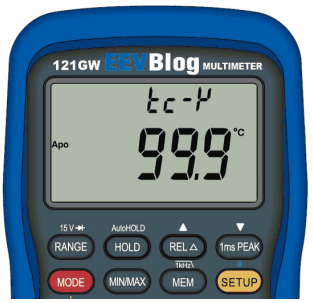


1 V = 2.218 dBm (600 Ω)



Choosing the reference

Six thermocouple types



Type K selected.

Where: temperature · settings page `tc`

Choose the type printed on your thermocouple. A wrong type gives a wrong temperature (a J sensor at 100 °C reads about 122 °C on the K setting).

Type	Range
K (default)	–200 to 1350 °C
J	–200 to 936 °C
T	–200 to 400 °C
N	–200 to 1300 °C
R, S	–50 to 1760 °C

Outside the range the meter shows `OFL`.

PT100 / PT1000 sensors



A PT100 at 100 °C.

Where: resistance · settings page `Pt`

Connect a PT100 or PT1000 sensor and read the temperature directly.

1. Settings page `PtOFF`: choose `Pt100` or `P1000`, save.
2. For the best result: press **RANGE** to the 500 Ω range (PT100) or 5 kΩ (PT1000), short the probes, press **REL**, then connect the sensor.

Range –200 to 850 °C. Logging and Bluetooth still record the resistance.

4–20 mA loop



12 mA = 50 %.

Where: dial A / mA, DC mA (lead in the mA socket) · settings page `LP`

Shows a loop current as a percentage: 4 mA = 0 %, 12 mA = 50 %, 20 mA = 100 %. A broken loop (0 mA) shows –25 %. Choose `L4–20`, or `L0–20` for 0 mA = 0 % and 20 mA = 100 %, and save. Logging and Bluetooth still record the current.

Better continuity

Where: dial Ω, continuity · threshold page and page `bH`

Threshold: `DN` = beep below the value (finding connections), `UP` = beep above the value (finding broken wires). Values 1, 5, 10, 30, 50, 100, 200 and 300 Ω. Default `DN 30`.

Beep hold (`bH`): how long the beep lasts after a very short contact, so you hear even a quick touch: 100, 50 or 25 ms, or `bHOFF` (shortest, about 15 ms).



Beep above 300 Ω.



Beep hold 25 ms.

Faster digits (display level)

Where: any function · settings page `dSP`

`dSP 1` = calm, steady digits (as the original). `dSP 2` = the digits follow changes about four times faster, for adjusting things or watching a value move. It does not change how often the display updates, in any function: that is set by the measuring chip (the 50 mV range, for example, updates more slowly than the 500 mV range). Only small changes are averaged, so in mV, where already a change of a few digits shows at once, dSP is hardly visible. Hz, ms and % are not affected.



Level 2.

Steady capacitance range (anti-hunting)

Where: capacitance · settings page `HUnt` (sub display), value on the main display

With some small ceramic capacitors of about 100 nF the capacitance range keeps changing between 100 nF and 1 µF: the digits stay blank and only the decimal point moves. The page `HUnt` has two values, shown on the main display: `oFF` (the default, as the original) and `on`. With `on` the meter stays on the 1 µF range after the 100 nF range has given no result twice in a row, and shows the value (about 0.095 µF for a 95 nF capacitor, one digit less than the 100 nF range would show). It goes back to normal ranging when a result arrives on the 100 nF range, when you connect a different capacitor, with open probes, when you change the range by hand, press REL or change the function, and when you switch the meter off. Leave it `oFF` if your capacitors measure normally. It does nothing in the other functions.



HUnt on.

Diode contact chirp



CHIrP, off.

Where: diode · settings page `CHIrP` (sub display), value on the main display

With `on` the diode double beep sounds once per diode contact, and not while it stays connected, with open probes, a reversed diode or when you lift the probes. `oFF` (the default) keeps the beep as before. With the buzzer page set to `b–oFF` the chirp is silent too; with `qUiEt` it sounds while the keys stay silent. Idea: GitHub user Cecilius.

Backlight timer



30 seconds.

Where: settings page `bL`

The light switches itself off after the time you choose: `bL30S` (30 seconds) or `bL 1` to `bL 99` (minutes). `bLOFF` = no timer (default). Any key or dial turn switches the light back on; that key press only wakes the light and does nothing else.

Faster Bluetooth



20 per second.

Where: settings page `bt`

How many readings per second go out: 2 (as the original), 4, 10 or 20. Use 10 or 20 for live graphs, 2 to save battery.

12-hour clock



18:25 shown as 6-25.

Where: settings page `CL`

`CL 12` shows the time as 1–12 (14:30 shows as 2-30; there is no AM/PM mark). Log files and Bluetooth keep 24-hour time.

Smart logging (trigger)

Where: settings pages `tr` and `Ch`

- `tr-In`: log at every interval (as before).
- `tr-Ch`: log only when the reading **changes** by at least the amount on page `Ch` (1 to 255 steps of the last digit). Perfect for long recordings of something that rarely moves.
- `tr-Lt`: log when the pass/fail result changes **page 10**.

With interval 0 the meter logs only on the trigger. With a longer interval it also logs at every interval (with `tr-Lt` only while the result is not PASS).



Trigger on change.



Change amount.

Battery check and trim

Where: settings page `BAT`

The low-battery symbol comes at 4.1 V. Some meters read the battery a few percent high or low. To correct it, hold **SETUP** on this page: `bA` and the voltage blink, REL / PEAK change it in small steps (up to ±10 %), hold SETUP to save.



Adjusting the battery reading.

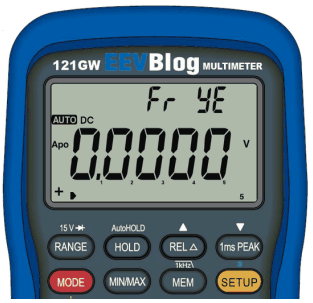
Quiet mode that is really quiet

The buzzer page has three settings. `b-On`: every beep. `b-OFF`: the key beeps and info beeps are silent. `qUiEt`: the keys are silent, but the meter still beeps for what it measures: the diode beeps, the auto-hold capture, the limit test, the auto power-off warning. In all three the continuity beep and the safety warnings (wrong socket, high voltage) sound.

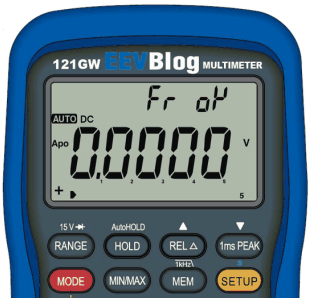
Faster measuring

The display updates more often than with the original firmware. In tests: currents, continuity and diode every 0.1 s, DC volts twice as often as before, and the first reading about half a second after switching on.

Big SD cards and formatting



Fr yE: format



Fr oK: done

The meter reads and writes cards formatted as FAT32 (usual for cards up to 32 GB) and **exFAT** (usual for larger cards), with long file names. With the two-part bootloader (**page 24**) you can also **format the card on the meter** (it becomes exFAT):

1. Settings page `Fr no` (after `Ld-bt`).
2. Hold SETUP, REL for `Fr yE`, hold SETUP. **Everything on the card is erased.**
3. `Fr --` while working, then `Fr oK`.

Or from the computer: with the meter on page `Ld-bt`, the Companion's CARD view has **Format card**. Type FORMAT to confirm; a window stays open until the meter is done.

Bluetooth file transfer



Ready for the app.



Transfer done.

Where: settings page `Ld-bt`, with the Companion app

With Bluetooth on and the page `Ld-bt` shown, the Companion app can send new firmware to the meter, list, download and delete the files on the SD card, and read and change most settings. Readings pause while the app uses the card. During a firmware transfer the meter shows the progress in percent (`Ld 42`) and `Ld oK` when done. Logging must be off. Speed: about 1 KB per second, so a firmware file takes about 2½ minutes.

Extended link diagnosis

Where: Companion app, SETTINGS view · firmware F-1.15 or newer

For a problem report about damaged data on the Bluetooth link. When it is on, the meter counts what happens on its Bluetooth serial line: bytes received, bytes lost, noise, framing errors, rejected commands and answers it could not send. The numbers show whether the damage happens in the meter, on the radio or in the computer.

1. In the Companion, open the SETTINGS view (key 6) and switch **Extended diagnosis** on. It is **off by default**.

2. Use the meter as usual; the Companion tells the meter to count when it connects.
3. On the METER view click **Save report**: the counts are in the report file.

The counters live only in the meter's RAM: they start at zero each time counting is switched on, are lost when you switch the meter off and are never stored. Counting changes nothing else on the link.

With firmware F-1.17 or newer the report also has a section **meter clock**. Each time the meter is switched on or wakes up, it measures its own run clock against its crystal, trims the clock and sets the speed of its Bluetooth serial line from the measurement. The section shows the clock as found, the clock after the trim and the resulting serial speed. Nothing needs to be set and nothing is stored.

Protected calibration backup

A calibration backup (cal.bin) can only be restored into the meter it came from. Another meter's file is refused and nothing changes. Holding MEM a bit too long while switching on no longer starts logging instead of the backup menu. page 22

The settings menu (SETUP)

All settings live on the small display. Press SETUP to go to the next page. The pages come in two groups: the **measurement pages**, which a SETUP press goes round, and the **meter pages** (clock, date, backlight, Bluetooth and the like), which open when you hold SETUP on the last measurement page, `SEt`. After the last meter page, a function change or switching off, SETUP starts with the measurement pages again.

Changing a setting

1. Press SETUP until the page shows.
2. Hold SETUP: the value blinks.
3. REL = up / next, PEAK = down / previous.
4. Hold SETUP to **save** (long beep). A short SETUP leaves without saving.

Date and time are saved in two steps: on the month-day page the month blinks first; hold SETUP, then the day blinks; hold SETUP again. The same for hour and minute. **Serial number**: a short SETUP moves to the next digit, REL / PEAK change it, hold SETUP to save.

Pages that only fit some functions (for example the thermocouple type) show only there. Saved settings stay after switching off. A page left blinking for about two minutes goes back to its saved value. The menu is locked while logging and while you look at stored readings.

All pages, in order

Page	What it sets	Where	
Measurement pages (press SETUP)			
`25.0c`	Temperature inside the meter; °C (`c`) or °F (`F`)	all	
extra pages	Depending on the function: frequency, dBm, continuity threshold, diode voltage, burden voltage	some	
`rEL-d`	REL as difference or percent	all	page 10
`tc-K`	Thermocouple type	temperature	page 12
`dSP 1`	Display level	all	page 13
`bH100`	Continuity beep hold	continuity	page 13
`LPOFF`	4–20 mA loop view	DC mA	page 12
`PtOFF`	PT100 / PT1000 view	resistance	page 12
`HUnt`	Capacitance anti-hunting (on / oFF on the main display)	capacitance	page 13
`CHIrP`	Diode contact chirp (on / oFF on the main display)	diode	page 14
`LtOFF` , `Lo` , `HI`	Pass / fail test and its limits	all	page 10
`PASS`	Pass / fail result	while the test is on	page 10
`tr-In` , `Ch`	Logging trigger and change amount (Ch only with tr-Ch)	all	page 15
`In 0`	Logging interval, 0–999 s (0 = about 5 per second)	all	page 19
`n` , `Sd` , `tHI` , `tLo`	Statistics	during MIN/MAX	page 11
`SEt`	Hold **SETUP** here for the meter pages	all	
Meter pages (hold SETUP on SEt)			
`BAT5.0`	Battery voltage, trim	all	page 15
`APO.30`	Auto power-off: off, 1, 15, 30, 60 min	all	
`b-On`	Buzzer on / off / quiet keys (qUiEt)	all	page 15
`Lcd-4`	Display contrast	all	
`2026` , `11-05` , `14-30`	Year; month and day; hour and minute	all	

Page	What it sets	Where	
`00000`	Serial number (0–99999)	all	page 22
`bLOFF`	Backlight timer	all	page 14
`bt 2`	Bluetooth speed	all	page 14
`Ld-bt`	Bluetooth transfer	all	page 16
`Fr no`	Format the SD card	two-part bootloader only	page 16
`CL 24`	12 / 24-hour clock	all	page 14



Auto power-off time



Buzzer



Contrast



Year



Logging interval



Continuity threshold

Logging and the SD card

Record readings



Logging.

1. Put a micro SD card in the slot under the battery cover.
2. Set the interval (page `In`) and, if you like, a trigger page 15.
3. Hold **MEM**: logging starts, MEM blinks, the small display counts.
4. Hold **MEM** again to stop.

During logging most keys are locked so nothing changes by accident; backlight and Bluetooth still work. With low batteries logging won't start or stops (`LoBAt`).

> **TIP** Stop logging before you switch off or take the card out, so the last readings are saved.

The files

Each recording is a file named after the date, for example **26110500.csv** (2026-11-05, recording 00). Open it in any spreadsheet: date, time, function, value and unit on each line, and at the end the maximum and minimum with the time they happened. A file holds up to 600 000 lines; then logging stops by itself (`SdEnd`).

Look at the last readings

Press **MEM**: the last logged readings (up to 60). **REL** = next, **PEAK** = previous, **MEM** = back. It closes by itself after about 10 seconds.

Cards

Use a micro SD card formatted as FAT32 or exFAT. Note: EEVblog's original bootloader can only read FAT32 cards for firmware updates.

Bluetooth and the Companion app

Hold **PEAK** to switch Bluetooth on (BT symbol); hold again to switch off. The meter keeps measuring and sends its readings. Only one device at a time: close the phone app before using the computer app.



Bluetooth on (BT).

Connecting: do not add (pair) the meter in the Bluetooth settings of Windows, macOS or Linux. The Companion finds the meter by itself; a meter that the computer has paired is kept connected by the computer and can no longer be found. If you added it already, remove it there, switch the meter's Bluetooth off and on (hold **PEAK** twice) and start the Companion. After an update over Bluetooth the meter switches Bluetooth on by itself (bootloader 1.1.4 or newer); with an older bootloader hold **PEAK**, and the Companion reconnects.

The **121GW Companion** is an app for Windows, macOS and Linux (download it from the project's releases page; free for individuals, schools and non-profits). It can:

- show the meter live (**LIVE** view) with statistics, and press the meter's keys from the computer;
- draw the readings of the last minutes on a scope grid (**SCOPE** view): choose the time span and the vertical scale, freeze the picture, and read time, voltage and frequency differences between two cursors;
- record to your computer (**LOG** view): start and stop a recording, see the recordings in the computer's folder, and save the trend as a CSV file;
- list, download and delete the card files, copy new log files to a folder, and format the card (**CARD** view). With firmware F-1.17 or newer the button **Leave Ld-bt** takes the meter off page `Ld-bt` (the next SETUP page, as with a SETUP press), so the readings come back without touching the meter;
- show the meter's versions, set its clock from a time server, update the firmware and the bootloader, and change the meter's settings (**METER** view), with the meter on page `Ld-bt`. Not the logging interval, the serial number or the calibration. The capacitance page `HUnt` (firmware F-1.16 or newer) and the diode page `CHIrP` (F-1.21 or newer) are among them;
- keep the program's own settings (**SETTINGS** view, key `6`): the folders for recordings and for the card copy, the CSV format (decimal point with comma-separated values, or decimal comma with semicolons), the time server and whether the meter's clock is set after each connection, how often to look for new firmware (at each connection, daily or never), diagnostic logging, the extended diagnosis **page 16**, and a reset of all these settings.

The keys `1` to `6` on the computer switch between the six views.

After a transfer (an update or a card file) the Companion shows one line about it: how many packets in how many seconds, how many were sent again, missing answers, damaged packets, and the slowest answer. It stays on the METER view as "last transfer". If something does not work, click **Save report** on the METER view: it writes one text file with versions, link details, the last transfer and the settings, anonymised, to attach to a problem report on GitHub.

Backup of your calibration

Your meter's calibration, serial number and settings live inside the meter. Keep a backup on the SD card and on your computer, in case something ever goes wrong.

Make a backup



SdCAL

1. Card in, dial to OFF.
2. Hold **MEM** and switch on. Let go. You see `SdCAL`.
3. Within 5 seconds press **REL**: `SdSAv`.
4. Within 5 seconds press **MODE**. A moment later: `SAvEd`. Switch off.

The card now has the file **cal.bin**: your meter's calibration, serial number and settings. Copy it to your computer and keep it safe. `-Err-` means it didn't work (card missing or faulty). Too slow? The meter just starts normally; switch off and try again.



SAvEd

Restore it

The same, but press **PEAK** instead of REL (`SdSEt`), then **MODE**. Only the backup of **this** meter is accepted; anything else shows `-Err-` and nothing changes.

> **WARNING** Restoring replaces the calibration, the serial number and all settings with those in the file.

Serial number

The serial number is a settings page. The backup is tied to it: **don't change the serial number**, or make a new backup right after.

Keys while switching on

Hold while switching on	What happens
MEM	Backup / restore menu (above)
MEM + **HOLD**	Firmware update from the card page 23

Hold while switching on	What happens
RANGE	All segments on (display test)
MODE	`CAL1`: switch off again. Pressing further keys would start calibration mode, which can ruin the calibration.

Updating the firmware

Updates keep your calibration, serial number and settings. Still: make a backup first and use fresh batteries.

Which bootloader do you have? (The bootloader is the small program that installs firmware.) If the settings menu has the page `Fr no` after `Ld-bt`, you have the **two-part bootloader**; otherwise EEVblog's original.

The easy way: Companion app

Bluetooth on, settings page `Ld-bt`. When the app offers an update, click it. The firmware goes to the card (`Ld 42` … `Ld oK`).

With the two-part bootloader the app then restarts the meter and installs it; you don't press anything. With EEVblog's bootloader you finish with the card steps below.



Update mode (-IAP-).

With an SD card

1. Copy the firmware as **EEVBlog.bin** into the top folder of the card (FAT32 for EEVblog's bootloader). Make sure the copy is complete: EEVblog's bootloader doesn't check the file.
2. Dial to OFF. Hold MEM and HOLD and switch on.
3. IAP (update mode) shows (`-IAP-` with the two-part bootloader). Press SETUP, within about 8 seconds (EEVblog's) or 30 seconds (two-part).
4. Wait until the meter restarts by itself. Don't take the card or batteries out before: EEVblog's bootloader erases the old firmware first, so an interrupted update leaves the meter without firmware until you repeat it with a good file. The two-part bootloader shows `douun`, then `Good`; `FAIL` = file refused, nothing changed.

The two-part bootloader

An optional replacement for EEVblog's bootloader. With it, updates go over Bluetooth without card or keys, exFAT cards work, and you can format the card on the meter. It never touches your calibration. The Companion app installs it once (fresh batteries needed, at least 4.8 V). More in the bootloader guide on the project page.

Messages

Display	Meaning	What to do
`OFL`	Input too big for the range or the sensor	Higher range or smaller input
`LEAd` + beeping	A lead is in a current socket, the dial is on something else	Move the lead to V Ω, or turn the dial to current
`-----`	No value yet	Wait or measure
`-oFF-`	Auto power-off	Press any key or turn the dial
`LoBAt`	Batteries too low for logging	New batteries
`SdEnd`	The log file is full	Start a new recording
`SdE03` etc.	SD card problem while logging	Check the card, its space and that it sits right
`SdCAL`, `SdSAv`, `SdSEt`	Backup menu	page 22
`SAvEd`	Backup / restore done	Switch off
`-Err-`	Backup / restore failed (no card, wrong file, other meter's file)	Check card and file
`Ld 42`, `Ld oK`	Bluetooth transfer running / done	–
`LdE 7`	Bootloader can't be installed on this meter	Nothing was changed
`LdE 8`	Batteries too low for the bootloader install	Fresh batteries
`LdE` other numbers	Transfer failed	Send again
`Fr oK`, `FrE n`, `FrE L`	Format done / failed / not now (logging or calibration mode)	Other card / stop logging
`-IAP-`	Update mode (two-part bootloader)	Send the firmware, or use a card
`douun`, `Good`, `FAIL`	Card update running / done / file refused	FAIL: check EEVBlog.bin
`CAL1`, `CAL`	Calibration menu / mode	Switch off

Troubleshooting

Problem	Solution
A key does nothing, two short beeps	Not possible right now, e.g. during logging. Stop logging first.
No `SdCAL`, or MODE does nothing after `SdSAv`	Hold MEM while switching on, then press each next key within 5 seconds.
Bluetooth won't connect	Hold PEAK until BT shows. Close the phone app: only one device at a time.
The range keeps jumping	Press RANGE to fix the range.
Faint display or ghost segments	Change the contrast (page `Lcd`).
The light goes off by itself	Page `bL`: `bL0FF`.
The meter switches itself off	Page `APOoF`.
Card update does nothing (EEVblog's bootloader)	Card must be FAT32, file EEVBlog.bin in the top folder.
`-IAP-` stays after an interrupted update	Send the firmware again with the app or use a card.
Display blank after a bootloader update	Take the batteries out and put them back.

Measuring ranges

The accuracy of each range is in EEVblog's original 121GW manual; the firmware doesn't change the hardware.

Function	Ranges
DC and AC volts	5 V, 50 V, 500 V, 600 V
DC and AC millivolts	50 mV, 500 mV
Low Z	600 V
Resistance	50 Ω, 500 Ω, 5 kΩ, 50 kΩ, 500 kΩ, 5 MΩ, 50 MΩ
Continuity	beep threshold 1–300 Ω
Diode	3 V or 15 V test voltage
Capacitance	10 nF to 10 000 µF

Function	Ranges
Frequency	99.999 Hz to 999.99 kHz
Pulse width	9.9999 ms, 99.999 ms, 999.99 ms
Duty cycle	to 99 %
DC and AC current	50 µA to 10 A
Power	250.00 µVA to 500.00 VA (50 V × 10 A)
Temperature	Type K –200 to 1350 °C, other types page 12